

Paper Summary

BalanceBenchmark: A Survey for Multimodal Imbalance Learning

English Summary with Figures and Tables

Core Topic

A survey and benchmark for evaluating multimodal imbalance learning methods using unified datasets, metrics, and experimental workflow.

Item	Details
Paper Type	Survey / Benchmark / Toolkit
Main Problem	Multimodal imbalance, where one modality dominates while others are underutilized.
Main Contribution	BalanceBenchmark and BalanceMM for fair comparison of multimodal imbalance algorithms.
Method Groups	Data, Feed-forward, Objective, and Optimization.
Why It Matters for Our Project	It helps justify that an audio-face prototype needs relative modality balance, not blind equal weighting.

1. Paper Overview

This paper surveys the multimodal imbalance learning problem and introduces BalanceBenchmark, a unified benchmark for comparing existing imbalance mitigation algorithms. The authors argue that many multimodal systems over-rely on a dominant modality, while weaker modalities are underutilized during training.

The paper is important because it does not focus on a single algorithm only. Instead, it organizes previous methods, compares them fairly, and provides a toolkit called BalanceMM to standardize experiments.

Main Idea

The paper shows that solving multimodal imbalance requires considering performance, imbalance degree, and computational complexity at the same time.

2. Problem Statement

Although many methods have been proposed to reduce multimodal imbalance, comparing them is difficult because prior studies often use different datasets, different metrics, and different training workflows.

- Lack of diverse and representative datasets for evaluating generalization.
- Lack of diverse evaluation metrics beyond accuracy or F1-score.
- Lack of standardized experimental workflow, which makes fair comparison unreliable.
- Existing methods may improve balance but still increase computational cost or reduce final performance.

3. Proposed Benchmark and Toolkit

To address these limitations, the paper introduces BalanceBenchmark and BalanceMM.

3.1 BalanceBenchmark

BalanceBenchmark is a comprehensive evaluation framework for multimodal imbalance algorithms. It includes multiple datasets, several modality combinations, and metrics from three perspectives: performance, imbalance degree, and computational complexity.

3.2 BalanceMM

BalanceMM is a modular toolkit that supports standardized experiments for different algorithms, datasets, backbones, and modality settings. It enables more reliable and reproducible comparisons.

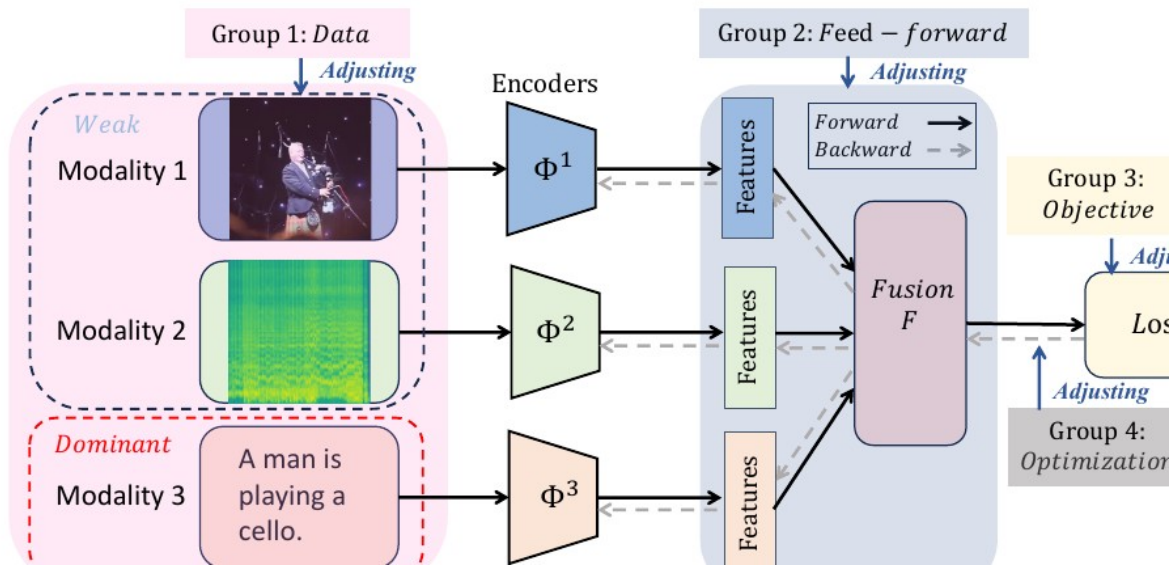


Figure 1. General framework and taxonomy of multimodal imbalance learning methods.

4. Taxonomy of Multimodal Imbalance Methods

The paper categorizes existing imbalance mitigation methods into four main groups based on where the adjustment is applied in the learning pipeline.

Category	Where It Acts	Main Idea	Example Methods
Data	Data processing	Adjust or resample data to enhance weak modality learning.	Modality valuation
Feed-forward	Feature and fusion process	Modify feature processing or fusion modules to reduce dominance.	MLA, OPM, Greedy, AMCo
Objective	Loss functions	Change learning objectives to promote weak modalities.	MMCosine, UMT, MBSO, CML, GBlending, MMPareto
Optimization	Training updates	Modify gradients or optimization behavior to balance modalities.	OGM, AGM, PMR, Relearning, ReconBoost

Table 1: Multimodal imbalance algorithms. **Adjustment Strategy** refers to different groups of methods in Section 3. **Imbalance Indicator** denotes the metric used to evaluate modality performance. **Number of Modalities** indicates the maximum number of modalities included in the experiments of the corresponding paper. **Dataset Domain** refers to the types of modalities included in the corresponding paper.

Method	Adjustment Strategy	Imbalance Indicator	Number of Modalities	Dataset Domain		
				CV	NLP	Audio
Modality-valuation [Wei <i>et al.</i> , 2024a]	Data	Shapley-based Metric	2	✓		✓
MLA [Zhang <i>et al.</i> , 2024]	Feed-forward	N/A	3	✓	✓	✓
OPM [Wei <i>et al.</i> , 2025]	Feed-forward	Performance Score	3	✓	✓	✓
Greedy [Wu <i>et al.</i> , 2022]	Feed-forward	Gradient Change	2	✓		✓
AMCo [Zhou <i>et al.</i> , 2023]	Feed-forward	Performance Score	3	✓	✓	✓
MMCosine [Xu <i>et al.</i> , 2023]	Objective	Performance Score	2	✓		✓
UMT [Du <i>et al.</i> , 2023]	Objective	N/A	3	✓		✓
MBSD [Liu <i>et al.</i> , 2023]	Objective	Performance Score	2	✓	✓	
CML [Ma <i>et al.</i> , 2023]	Objective	Classification Loss	2	✓	✓	
MMPareto [Wei and Hu, 2024]	Objective	Performance Score	3	✓	✓	✓
GBLending [Wang <i>et al.</i> , 2020]	Objective	Classification Loss	3	✓		✓
LFM [Yang <i>et al.</i> , 2024]	Objective	N/A	3	✓	✓	✓
OGM [Peng <i>et al.</i> , 2022b]	Optimization	Performance Score	2	✓		✓
AGM [Li <i>et al.</i> , 2023]	Optimization	Performance Score	3	✓	✓	✓
PMR [Fan <i>et al.</i> , 2023]	Optimization	Prototype	2	✓		✓
Relearning [Wei <i>et al.</i> , 2024b]	Optimization	Clustering	3	✓	✓	✓
ReconBoost [Hua <i>et al.</i> , 2024]	Optimization	Classification Loss	3	✓	✓	✓

Table 1. Taxonomy of multimodal imbalance algorithms, their strategies, indicators, and modality domains.

5. Evaluation Metrics

The paper evaluates methods using three main perspectives:

- Performance: Top-1 Accuracy and F1-score.
- Imbalance Degree: a Shapley-value-based metric that estimates modality contribution.
- Complexity: FLOPs, which measure the computational cost required for training.

Observation from Figure 2	Meaning
Multimodal model achieves higher overall performance than individual modalities.	Combining modalities can improve final prediction.
Audio and video inside the multimodal model can perform worse than audio-only or video-only models.	Some modalities are suppressed during joint multimodal training.
The weaker modality shows a larger performance gap.	This confirms the multimodal imbalance phenomenon.

6. Main Experimental Insights

The experiments show that most imbalance mitigation methods improve performance over the baseline, confirming that multimodal imbalance is a real and common issue. However, the paper also finds that no existing method achieves the best balance between performance, modality balance, and computational cost.

Method Group	Main Strength	Main Limitation
Objective-based	Performs well on many datasets and improves weak modality learning.	May not work best when the imbalance degree is already small.
Optimization-based	Provides fine-grained control over training and works well for small imbalance.	Highest computational cost.
Feed-forward-based	Lowest average FLOPs and relatively low overhead.	May be less general because it depends on specific feature/fusion designs.
Data-level	Improves training data quality before learning.	Can introduce relatively high computational cost.

7. Relative Balance vs. Absolute Balance

One of the strongest insights in the paper is that absolute balance does not always lead to better performance. Different modalities naturally carry different amounts of information. Therefore, blindly forcing all modalities to contribute equally may reduce performance by weakening high-information modalities.

The paper suggests that relative balance is more useful than absolute balance. Relative balance means preventing one modality from dominating while still allowing more informative modalities to contribute more when appropriate.

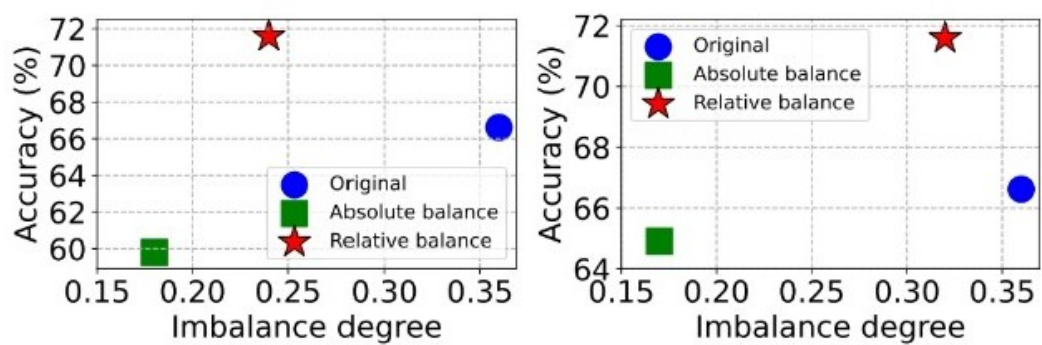


Figure 3. Relative balance can outperform absolute balance in terms of final accuracy.

8. Computational Cost

The paper compares the average FLOPs of different method categories. Optimization-based methods have the highest average computational cost, while feed-forward methods have the lowest.

Category	Average FLOPs (x10 ¹³)	Interpretation
Objective	8.38	Moderate computational cost.
Optimization	16.90	Highest cost due to fine-grained training adjustments.
Forward	6.94	Lowest cost because changes are mainly in feature/fusion flow.
Data	11.40	Relatively high cost due to data-level adjustments.

9. Research Gap for Our Project

Research Gap

BalanceBenchmark shows that existing multimodal imbalance methods still do not achieve a satisfactory trade-off between performance, modality balance, and computational cost. Moreover, pursuing absolute balance does not always improve performance. This creates a practical gap for real-world multimodal CIR prototypes, where audio and facial-expression modalities may differ in quality, reliability, and computational cost.

For our proposed prototype, which focuses on audio and facial expressions, this paper supports the idea that the system should avoid over-relying on one modality. However, the model should also avoid forcing equal contribution from both modalities when one of them is more informative or more reliable in a specific context.

10. Key Takeaways

- Multimodal imbalance is a major challenge in multimodal learning.
- Existing methods can be categorized into Data, Feed-forward, Objective, and Optimization strategies.
- BalanceBenchmark provides a fair and standardized way to compare these methods.
- No existing method perfectly balances performance, modality balance, and computational cost.
- Relative balance is more practical than absolute balance for real-world systems.
- This paper strongly supports the need for a reliability-aware audio-visual prototype.

11. Recommended Figures and Tables for Project Use

- Figure 1 - General Framework and Taxonomy: explains where each imbalance strategy acts.
- Figure 2 - Performance Gap: shows how modalities can be suppressed inside multimodal training.
- Figure 3 - Relative vs. Absolute Balance: supports the idea that relative balance is more practical.
- Table 1 - Methods Taxonomy: summarizes the major imbalance algorithms and their categories.
- Table 3 - FLOPs Comparison: supports the computational-cost discussion.